Moustafa Abdelwanis

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Google Scholar

WORK HISTORY

Khalifa University

Abu Dhabi, UAE
Jan 2021 - Dec 2026

Graduate Research and Teaching Assistant (Engineering Systems and Management)

- Conduct graduate-level research in Engineering Systems and Management, with a focus on operations management, AI in healthcare, and human-AI interaction.
- Support teaching activities in relevant courses, including tutorials, grading, and student supervision.
- Contribute to various research projects leading to peer-reviewed publications and conference presentations.

EnvoTech for Environmental and Ventilation Projects

Cairo, Egypt
Dec 2019 - Dec 2020

Technical Office and Design Engineer

- Designed dust collection filters for industrial applications.
- Prepared technical designs and documentation for environmental and ventilation projects.
- Supervised manufacturing and assembly processes to ensure compliance with specifications and quality standards.

QUALIFICATIONS

Khalifa University

Abu Dhab, UAE
2023 - 2026

PhD in Engineering

Khalifa University

Abu Dhabi, UAE
2021 - 2022

Master's in Engineering Systems and Management

American University in Cairo
Cairo, Egypt
2016

Bachelor of Science, Mechanical Engineering (specialization in Industrial Engineering)

SKILLS

- **Methods & tools:** Simulation modeling; Optimization; Variable neighbourhood search; Bowtie analysis; Structural Equation Modeling; SEIPS.
- **Programming / software:** Python; R; AnyLogic; Jamovi.
- **Languages:** Arabic; English

Select/Recent Publications

Abdelwanis, M., Simsekler, M. C. E., Gabor, A. F., Sleptchenko, A., Omar, M. (2026). *Artificial intelligence adoption challenges from healthcare providers' perspectives: A comprehensive review and future directions.* **Safety Science**, 193, 107028. DOI: 10.1016/j.ssci.2025.107028

Abdelwanis, M., Moawad, K., Mohammed, S., Hummieda, A., Syed, S., Maalouf, M., Jelinek, H. F. (2025). *Sequential classification approach for enhancing the assessment of cardiac autonomic neuropathy.* **Computers in Biology and Medicine**, 190, 109999. DOI: 10.1016/j.compbiomed.2025.109999

Abdelwanis, M., Gabor, A. F., Mladenovic, N., Sleptchenko, A. (2024). *Simulation-based variable neighborhood search for optimizing skill assignments in multi-server facilities with inventories.* **Computers & Operations Research**, 164, 106546. DOI: 10.1016/j.cor.2024.106546

Abdelwanis, M., Alarafati, H., Tammam, M., Simsekler, E. (2024). *Exploring the risks of automation bias in healthcare artificial intelligence applications: A Bowtie analysis.* **Journal of Safety Science and Resilience**. DOI: 10.1016/j.jnlssr.2024.06.001

Bader, M., **Abdelwanis, M.**, Maalouf, M. et al. (2024). *Detecting depression severity using weighted random forest and oxidative stress biomarkers.* **Scientific Reports**, 14, 16328. DOI: 10.1038/s41598-024-67251-y

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Conferences

- **Winter Simulation Conference (WSC)** – 2024
- **Computers and Industrial Engineering (CIE 50) – 2023**
- **International Conference on Variable Neighborhood Search (ICVNS) – 2022**

ACHIEVEMENTS

- Best Paper Award, 50th International Conference on Computers and Industrial Engineering (CIE 50), 2023.
- Best Paper Award, Graduate Student Research Competition (GSRC), American University in Sharjah, 2024.

REFEREES

Mecit Emre Simsekler
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Andrei Sleptchenko
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